

Week 2 Homework Assignment

Loop Control & Game Loops

Total Points: 100

Name: _____

Class: _____

Important Instructions

- Work on your own, but you may ask family members for hints only if stuck
- For setting up your code files, kindly see `instructions.pdf`
- For written answers, use the spaces provided on this sheet
- Show your work for full credit!

1 Part 1: Code Tracing with Loop Control (15 points)

Time estimate: 20 minutes

1.1 Exercise 1.1: Trace the Password Checker (8 points)

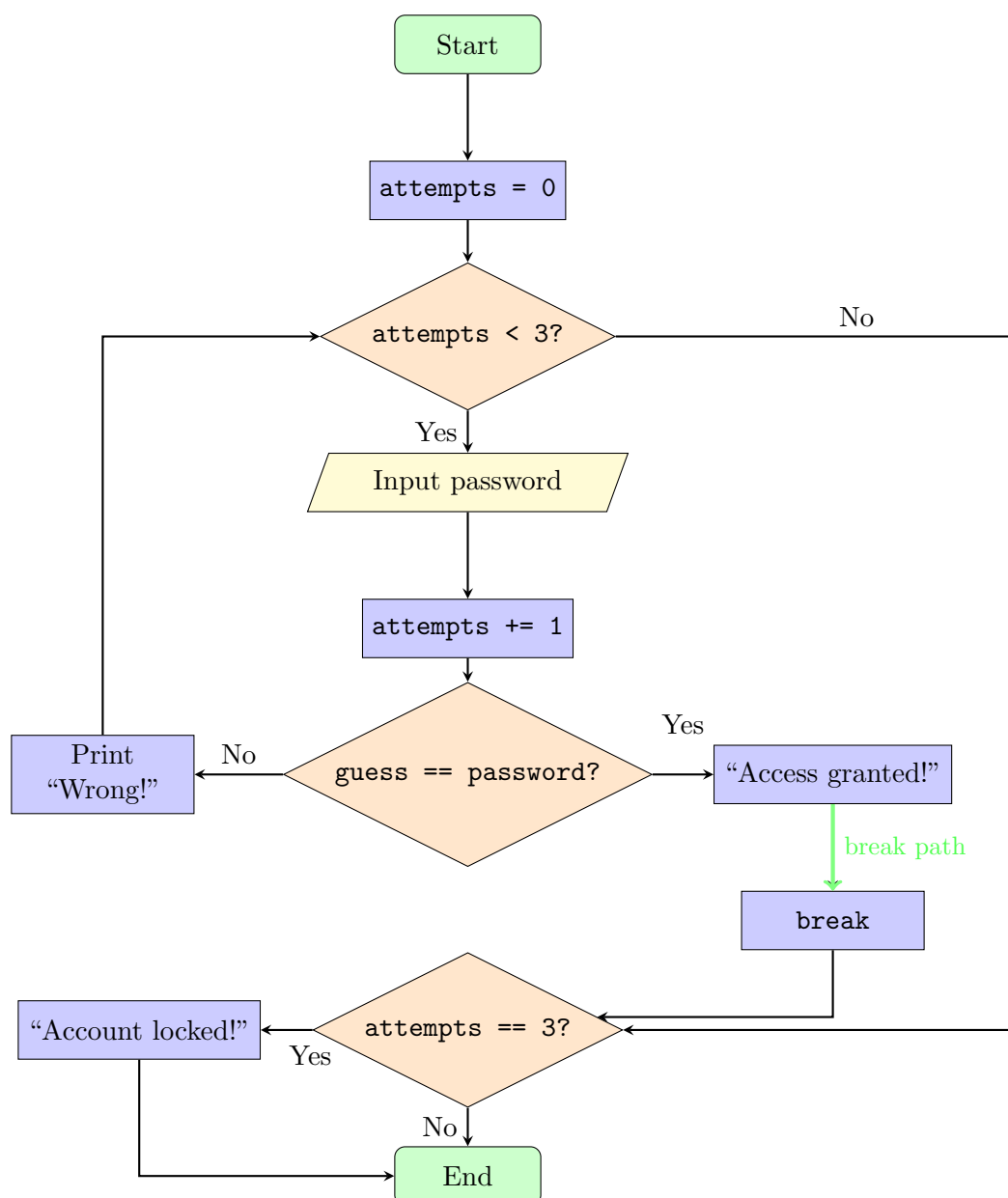
Read this code carefully and trace through each iteration:

```
attempts = 0
password = "python123"

while attempts < 3:
    guess = input("Enter password: ")
    attempts = attempts + 1

    if guess == password:
        print("Access granted!")
        break
    else:
        print(f"Wrong! {3 - attempts} tries left")

if attempts == 3:
    print("Account locked!")
```



Fill in what happens with these inputs:

Scenario A: User gets password correct on second try

Iteration	Input	attempts	Loop Condition	What prints
1	"hello"	1	1 < 3 (True)	Wrong! 2 tries left
2	"python123"			
After loop	–			

Scenario B: User uses all 3 attempts incorrectly

Iteration	Input	attempts	Loop Condition	What prints
1	"test"			
2	"password"			
3	"123456"			
After loop	–		attempts == 3?	

Scenario C: User gets it right on first try

Iteration	Input	attempts	Loop Condition	What prints
1	"python123"			
After loop	—			

1.2 Exercise 1.2: Continue Statement Trace (7 points)

What does this program print? Trace each iteration:

```
count = 0
while count < 5:
    count = count + 1
    if count == 3:
        continue
    print(f"Count is {count}")
print("Done!")
```

Your output prediction (one line per print):

2 Part 2: Code Completion - Loop Control (18 points)

Time estimate: 25 minutes

2.1 Exercise 2.1: Number Guessing Game (10 points)

Complete this guessing game that uses break to exit when correct:

```
import random
secret = random.randint(1, 10)
tries = 0

print("I'm thinking of a number between 1 and 10!")

while _____: # Loop forever
    tries = tries + 1
    guess = int(input("Your guess: "))

    if guess == secret:
        print(f"Correct! It took {tries} tries!")
        _____ # Exit the loop
    elif guess < secret:
        print("Too low!")
    else:
        _____ # What should print here?
```

Save this code as week2_ex2.1.py

2.2 Exercise 2.2: Skip Even Numbers (8 points)

Complete this code that prints only odd numbers using continue:

```
# Print odd numbers from 1 to 10
num = 0
while num < 10:
    num = _____ # Increment first
```

```
if num % 2 == 0: # If number is even
    ----- # Skip to next iteration

print(f"Odd number: {num}")
```

Save this code as `week2_ex2_2.py`

3 Part 3: Loop Control Practice (15 points)

Time estimate: 20 minutes

3.1 Exercise 3.1: Menu System (15 points)

Scenario: Pizza Order System

You're creating a simple pizza ordering system. The program should keep showing a menu until the user chooses to exit.

Complete this menu system:

```
print("=== Pizza Palace ===")
total = 0

while True:
    print("\n1. Small Pizza - Rs. 500")
    print("2. Medium Pizza - Rs. 800")
    print("3. Large Pizza - Rs. 1200")
    print("4. Exit and pay")

    choice = input("\nYour choice: ")

    if choice == "1":
        total = total + 500
        print("Small pizza added!")
    elif choice == "2":
        ----- # Add medium pizza price
        print("Medium pizza added!")
    elif choice == "3":
        ----- # Add large pizza price
        ----- # Print message
    elif choice == "4":
        print(f"Total bill: Rs. {total}")
        ----- # Exit the loop
    else:
        print("Invalid choice! Try again.")
        ----- # Skip to next iteration

print("Thank you for ordering!")
```

Save this code as `week2_ex3_1.py`

4 Part 4: Debug Detective (17 points)

Time estimate: 25 minutes

Each code snippet has errors. Find and fix ALL errors:

4.1 Snippet 1: Infinite Loop Bug (6 points)

```
# This code should count from 1 to 5 but has an infinite loop!
count = 1
while count <= 5:
    print(f"Count: {count}")
    # Oops! Forgot something important here
print("Finished counting!")
```

Fixed version:

```
# Write the corrected code here:
count = 1
while count <= 5:
    print(f"Count: {count}")
    ----- # What's missing?
print("Finished counting!")
```

Save your corrected code as `week2_ex4.1_fixed.py`

4.2 Snippet 2: Break Logic Error (6 points)

```
# This code should stop when user types "quit" but doesn't work
while True:
    user_input = input("Enter command (quit to exit): ")
    print(f"You said: {user_input}")
    if user_input == "Quit": # Bug here!
        break
```

What's wrong? -----

Save your corrected code as `week2_ex4.2_fixed.py`

4.3 Snippet 3: Continue Placement Error (5 points)

```
# Should print numbers except 5
num = 0
while num < 10:
    num = num + 1
    print(f"Number: {num}")
    if num == 5:
        continue # This doesn't skip printing 5!
```

Fix the code to actually skip printing 5:

Save your corrected code as `week2_ex4.3_fixed.py`

5 Part 5: Creative Challenge - Rock Paper Scissors (20 points)

Time estimate: 30 minutes

Create a Rock-Paper-Scissors game that:

- Keeps playing rounds until someone wins 3 times
- Uses break when someone reaches 3 wins
- Validates input (*only accept rock/paper/scissors*)
- Shows the score after each round

```

import random

print("=== Rock Paper Scissors - First to 3 Wins! ===")
player_score = 0
computer_score = 0

while True:
    # Get player choice
    player = input("\nRock, Paper, or Scissors? ").lower()

    # Validate input
    if player != "rock" and player != "paper" and player != "scissors":
        print("Invalid choice! Try again.")
        continue # Skip to next iteration

    # Computer choice (using random number instead of list)
    computer_num = random.randint(1, 3)
    if computer_num == 1:
        computer = "rock"
    elif computer_num == 2:
        computer = "paper"
    else:
        computer = "scissors"

    print(f"Computer chose: {computer}")

    # Determine winner (you complete this part)
    if player == computer:
        print("It's a tie!")
    elif (player == "rock" and computer == "scissors") or \
        (player == "paper" and computer == "rock") or \
        (player == "scissors" and computer == "paper"):
        # Update player score
        print("You win this round!")
    else:
        # Update computer score
        print("Computer wins this round!")

    # Show scores
    print(f"Score - You: {player_score}, Computer: {computer_score}")

    # Check for winner
    if player_score == 3:
        print("\nYou won the match!")
        # Exit loop
    elif computer_score == 3:
        print("\nComputer won the match!")
        # Exit loop

print("Thanks for playing!")

```

Save this code as `rock_paper_scissors.py`

6 Part 6: Problem Solving - ATM Simulator (10 points)

Time estimate: 20 minutes

6.1 Exercise 6.1: Build an ATM System (10 points)

Create a simple ATM program with these features: - PIN validation (max 3 attempts) - Menu with withdraw, deposit, check balance - Exit option - Use break for wrong PIN and exit

Starter structure:

```
correct_pin = "1234"
balance = 5000
attempts = 0

# PIN validation loop
while attempts < 3:
    # Your code here for PIN checking
    # Use break when PIN is correct
    pass

# If PIN was wrong 3 times, exit
if attempts == 3:
    print("Card blocked!")
else:
    # Main ATM menu loop
    while True:
        # Show menu and handle choices
        # Use break to exit
        pass
```

Save your complete program as *atm_simulator.py*

7 Part 7: Reflection Questions (5 points)

Time estimate: 10 minutes

Answer these questions in complete sentences:

1. When would you use **break** instead of making the while condition False? Give a specific example.

2. What's the difference between **break** and **continue**? How do they change program flow?

3. What debugging strategy helped you most with finding infinite loops?

8 Bonus Section: Extra Challenges (Optional - 10 extra points)

Only attempt if you've finished everything else!

8.1 Challenge 1: Advanced Menu System (5 points)

Enhance your pizza ordering system to: - Add drink options in a submenu - Calculate tax (10%) - Apply discount codes

Save as *week2_bonus1.py*

8.2 Challenge 2: Pattern Printer (5 points)

Create a program that prints this pattern using nested while loops and continue:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Skip printing even numbers!

Save as `week2_bonus2.py`

Submission Checklist

Before submitting, make sure you have:

- ☐ Completed all required sections (Parts 1-7)
- ☐ Saved each code exercise in its own file:
 - `week2_ex2_1.py`, `week2_ex2_2.py`, etc.
 - `rock_paper_scissors.py`
 - `atm_simulator.py`
- ☐ Created a folder named `week2_hw_yourname` with all files
- ☐ Tested each code file individually to ensure it runs without errors
- ☐ Written your name at the top of this paper
- ☐ Answered all reflection questions
- ☐ Attempted bonus section (optional)

Time Tracking:

How long did this homework take you? _____ hours

Parent/Guardian Signature: _____

(Confirming student completed work independently)

Great job mastering loop control!

Next week: Complex Loops & Patterns